

Towards real time crack detection as a source of ground truth

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2 November 2025

Abstract

Real-time crack detection is critical for maintaining the safety, reliability, and longevity of infrastructure systems such as bridges, tunnels, pavements, and buildings. Manual inspection remains the industry norm, yet it is labor-intensive, inconsistent, and difficult to scale. Recent advances in lightweight segmentation networks have opened the door to automated crack detection capable of operating at high resolution and near real-time performance. This work evaluates and refines state-of-the-art deep learning models for crack segmentation, with a particular focus on the HrSegNet architecture and the impact of loss-function design, transfer learning strategies, and training hyperparameters on real-world performance. Using the Crack Conglomerate Dataset and the LCW dataset as benchmarks, we examine four representative segmentation models—DeepLabV3+, YOLOv8-crack-seg, SCSegamba, and HrSegNetB32—across fidelity metrics (IoU, F1), speed (FPS), and robustness under super-resolution scaling factors. A detailed ablation study of HrSegNetB32 reveals that the commonly used OhemCrossEntropy loss is insufficient for highly imbalanced crack segmentation tasks, frequently collapsing to zero crack predictions. In contrast, the FocalTversky loss function yields substantial improvements, with transfer learning and increased training iterations further enhancing crack-level segmentation accuracy. Although segmentation performance on LCW remains challenging due to extreme variability and scarce crack pixels, this work demonstrates clear pathways toward stable, real-time crack detection and establishes a reproducible framework for future system-level deployment.

Introduction

The ability to predict crack generation in real time offers the ability to not only save money but also improve standards of safety. Concrete construction forms the backbone of infrastructure in the modern world due to its durability, cost effectiveness, and material availability [1]. Cracks in concrete typically originate on and are observable from the surface of the structure. Being an easy to observe and reliable metric of structural health and wear, cracks are an essential metric used in structural inspections [1]. Real-time crack detection would facilitate early warning of structural failures in dams, tunnels, buildings, and roads. Automating this process would reduce human error and enable non-intrusive monitoring even in dangerous or hard-to-reach areas. The added safety made possible by these technologies has the potential to reduce loss of life and property as well as increase public trust in the safety of aging infrastructure. Real-time segmentation would enable condition-based maintenance as opposed to routine or reactive repairs. Asset owners would be able to prioritize repairs, thereby optimizing maintenance schedules and budgets. The lowered need for manual inspections would reduce the labor cost associated with the upkeep of infrastructure. Increased accuracy in structural assessment would ensure that repairs are done when and where truly needed, preventing the waste of resources on pointless over-maintenance.

Related Work

Building on previous crack segmentation deep learning models, Liu et al. introduced SCSegamba, a lightweight structure-aware vision Mamba network for use in crack segmentation [6]. Using a Structure-Aware Visual State Space module SCSegamba is able to capture crack morphology and texture efficiently while requiring minimal computational resources [6]. SCSegamba integrates structure-awareness to capture both linear and irregular crack patterns [6]. CrackMamba integrates the Mamba architecture with attenuation mechanisms, enhancing the model's ability to capture global contextual information while still maintaining linear spatial and computational complexity [16]. When evaluated on datasets comprising concrete pavement, asphalt pavement, and steel cracks, CrackMamba was shown to consistently improve the baseline model's performance [16]. Improvements were shown across all evaluation metrics including both IoU and mIoU, while also reducing the number of parameters and computational costs [16]. CT-CrackSeg uses BAM to improve fine-edge detection as well as boundary sharpness [17]. CT-CrackSeg architecture supports domain transfer due to its hybrid convolution-transformer structure [17]. While the lightweight design offers greater efficiency than full transformer models, it is not yet sufficiently fast to be considered fully real time [17]. CrackNex integrates Retinex theory to separate reflectance from illumination, allowing the model to focus on crack features independent of lighting variations, addressing the issue of poor lighting which is one of the largest real-world challenges in structural health monitoring [18]. CrackNex is designed to perform effectively with a limited number of annotated samples using a support-query prototype learning framework, allowing for rapid adaptation to new environments [18]. By introducing a fusion model integrating both support and reflectance prototypes. CrackNex offers both increased segmentation accuracy and robustness [18]. CrackSegFormer uses Transformer-based architecture to capture multi-scale features, thereby improving the model's ability to detect cracks of various sizes and complexities [19]. CrackSegFormer enhances the model's understanding of context by focusing on semantic segmentation, allowing more accurate delineation of cracks from the surrounding surface [19]. CrackSegFormer has the added benefit of being adaptable to various surface types [19]. HrSegNet4CrackSegmentation maintains high resolution feature maps allowing for the detection of small and thin cracks [20]. The Semantic Guidance Module merges high-level semantic information into early, detail-rich layers – increasing the model's ability to distinguish between cracks and background noise [20]. The efficient architecture of HrSegNet4CrackSegmentation allows competitive speed to be achieved without sacrificing performance [20].

For my master's thesis, I had focused on building a crack segmentation dataset and wanted to compare different neural network models to see how mine would compare, ensuring that I build SOTA crack segmentation and width estimation ground truth.

Methods

For the HrSegNet loss functions, I compared OhemCrossEntropy loss, which the base HrSegNet models used for training, and FocalTversky loss function. OhemCrossEntropy picks out a subset of pixels with the highest training loss and only computes loss on them, helping the model focus its training on pixels it specifically is making mistakes on. The

OhemCrossEntropyLoss is defined as: $\frac{1}{H} \sum_{i \in H} l_i(Pred_i, Label_i)$, where H is the set of pixels with

the highest cross entropy loss, and the per-pixel cross entropy loss l_i is defined as

$$l_i = - [Label_i \log(Pred_i) + (1 - Label_i) \log(1 - Pred_i)].$$

FocalTversky combines the Focal exponent and the Tversky index to address the issue of class imbalance in training, with Tversky being a generalization of the dice coefficient with extra weights to balance false positives and negatives, and Focal making the loss focus on hard to predict pixels [8]. Focal Tversky loss is defined as $FTL = (1 - TI)^\gamma$, where γ controls the linearity of the loss, and $TI = \frac{TP}{TP + \alpha FN + \beta FP}$, where $\alpha + \beta = 1$, and if $\alpha = \beta = .5$, it simplifies to the Dice Coefficient, and $\alpha > \beta$ penalizes False Negatives more [8]. This helps the model learn to better segment thin, rare, or small crack regions without being overwhelmed by the easy, dominant background pixels.

For crack segmentation at the core of the pipeline, for the sake of expediency I used 100 image subset (seed of 42) from the Crack Conglomerate Dataset test set as a benchmark, as well as using bicubic as a simple, efficient super-resolution algorithm to compare the level of model performance degradation without the hurdle of implementing the dscv ai model into the system. I compared four different models: DeeplabV3+, trained on CCD, as the baseline, HrSegNet, trained on CrackSeg9k, as a specialized model for high resolution, YOLOv8-crack-seg, trained on the Crack Seg dataset, as a standard modern crack detection model, and SCSegamba, trained on TUT, as a new candidate model claiming to improve upon other models. For performance degradation, I compared mIoU, F1 Score, FPS, and the mIoU between original (1x), double resolution (2x), and quadruple resolution (4x), with the mIoU ratio of 2x/1x and 4x/1x being calculated for further clarity. I attempted to run CrackNex for testing, but faced memory spikes that prevented me from testing it even at 2x scale due to the model being built for small input sizes.

Selecting the HrSegnet as the best segmentation model, the HrSegNet models are trained using the PaddleSeg built in trainer. The base b32 models were downloaded from the [HrSegnet github repository](#), and were fine tuned and transfer learned in the same manner as they were trained on the Labeled Cracks in the Wild dataset, called via a shell or powershell script. For training, warmup iterations were set to 2000, batch size to 32, 100,000 iterations and a learning rate of .01. For the ablation study, hrsegnetb32 was chosen for its higher inference capability compared to hrsegnetb16 while not being as costly or demanding a large memory footprint like the larger hrsegnetb48.

Transfer learning experiments were done on the LCW dataset for the full dataset and noblanks subset for analysis across a wide range of parameters. For the model, B3B+ transfer learning is done on the model. For heavy transfer learning (Heavy TL), warmup iterations were set to 1000, learning rate to .01, and total iterations to 5000. For light transfer learning (Light TL), warmup iterations were set to 500, learning rate to .0001, batch size to 8 for 20000 iterations. Transfer Learning was done on an Alienware Aurora R13 desktop with an RTX 3090 and 128gb of RAM, and the models were evaluated on the LCW test dataset to evaluate its performance on difficult real world cracks.

Case Name	Warmup Iterations	Learning Rate	Iterations	Eval Increment	Batch Size
Light TL	500	0.0001	5000	500	8
Heavy TL	1000	0.01	20000	1000	8
Base	2000	0.01	100000	1000	32

Figure: Parameters for base model and transfer learning variants

I tested the original OhemCrossEntropy loss function for transfer learning and fine tuning, but found that the model was unable to predict any crack pixels due to the severe class imbalance between background and crack pixels.

As a result, further testing was done with the FocalTversky loss function, with more and less aggressive hyperparameters to account for the class imbalance being compared, with base parameters being $\alpha = 0.3, \beta = 0.7, \gamma = 1.0$, and more aggressive parameters being $\alpha = 0.2, \beta = 0.8, \gamma = 1.33$.

FocalTversky demonstrated dramatic improvements over OhemCrossEntropy for both parameter sets, with transfer learning demonstrating much greater improvements over fine tuning, but the accuracy was far below acceptable for the crack, so further refinements were experimented on. To continue refining, doubling the learning rate to .02 and doubling the iterations to 40,000 for transfer learning to see if an increased training time or learning rate would further help were tested. For reference, the creators of the dataset were only able to achieve 18.77% correctly predicted cracks when training on LCW testing on the LCW dataset, and 8.34% fine tuning on the entire LCW dataset from the original Crack Conglomerate dataset, which shares many of the same sub-datasets as CrackSeg9k. [3].

Training dataset	Target dataset	Correctly detected cracks (%)
Conglomerate	LCW	1.13
LCW	LCW	18.77
LCW (FT)	LCW	8.34
LCW (no blank)	LCW (no blank)	39.95
LCW (no blank) (FT)	LCW (no blank)	20.69
LCW (no blank)	Conglomerate	16.97
LCW (no blank) (FT)	Conglomerate	17.85
Conglomerate	Conglomerate	70.70

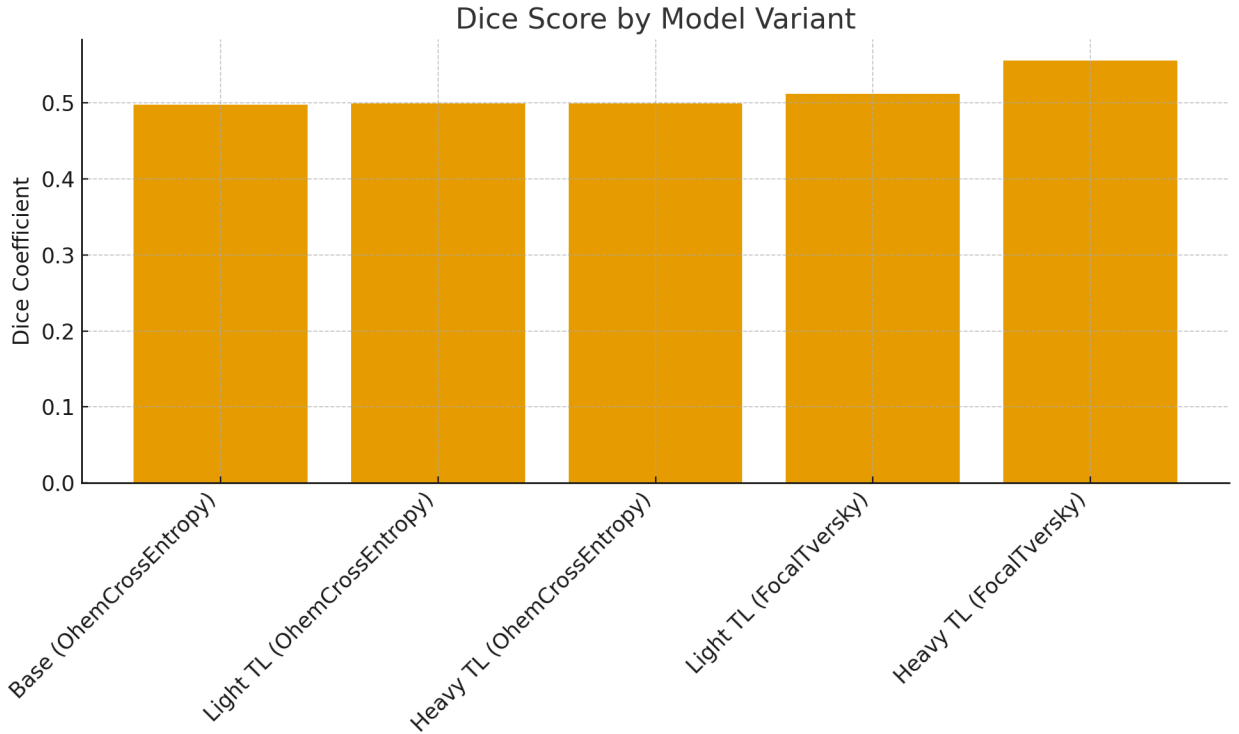
Figure: Segmentation results from LCW authors for DeepLabV3+ network
Results

For the ablation study, the HrSegNetB32 model trained on the CrackSeg9k dataset is referred to as Base, and transfer learned models as Heavy TL and Light TL. For parameter comparison, please see Table x. While the base model, as well as the transfer learned versions of the model with the OhemCrossEntropy loss function were unable to create any internal model of the crack structure of the cracks, as seen by the near-zero IoU scores for the crack for HrSegNet original, models utilizing the base less aggressive FocalTversky loss function

demonstrated dramatically improved, albeit highly inaccurate results at correctly segmenting crack pixels, with FocalTversky achieving a 6 percent IoU score for crack detection from 0 percent for the base, Light and Heavy TL learned models.

Model Variant	Loss Function	mIoU	IoU_bg	IoU_crack	Precision_bg	Precision_crack	Recall_bg	Recall_crack	F1_bg	F1_crack	mean_Precision	mean_Recall	mean_F1	FPS	Kappa	Dice
Base	OhemCrossEntropy	0.488702765	0.972443	0.004962969	0.997137	0.005519418	0.975165	0.04691802	0.986029	0.009877	0.501328205	0.51104167	0.497953	24.58	0.00466	0.497953
Light TL	OhemCrossEntropy	0.498535425	0.997071	0	0.9970708	0	1	0	0.998533	0	0.498535425	0.5	0.499267	24.27	0	0.499267
Heavy TL	OhemCrossEntropy	0.498535425	0.997071	0	0.9970708	0	1	0	0.998533	0	0.498535425	0.5	0.499267	23.54	0	0.499267
Light TL	FocalTversky	0.503658331	0.993093	0.014224138	0.9971586	0.023859631	0.995911	0.03402375	0.996534	0.028049	0.510509136	0.51496724	0.512292	23.94	0.024691	0.512292
Heavy TL	FocalTversky	0.528556947	0.996819	0.060295074	0.9972731	0.309425183	0.999543	0.06967037	0.998407	0.113733	0.653349154	0.53460679	0.55607	24.53	0.112777	0.55607

Table: comparison of base, fine tuned, transfer learned models and loss types



Further experiments comparing the learning rates and the number of iterations done on Heavy TL were done, increasing the learning rate and iterations each resulted in a higher mIoU than the original FocalTversky model, with increased iterations from 20000 to 40000 having the biggest impact on improving the mean IoU and other metrics, with 40,000 iterations having the highest crack IoU score of 7.49%.

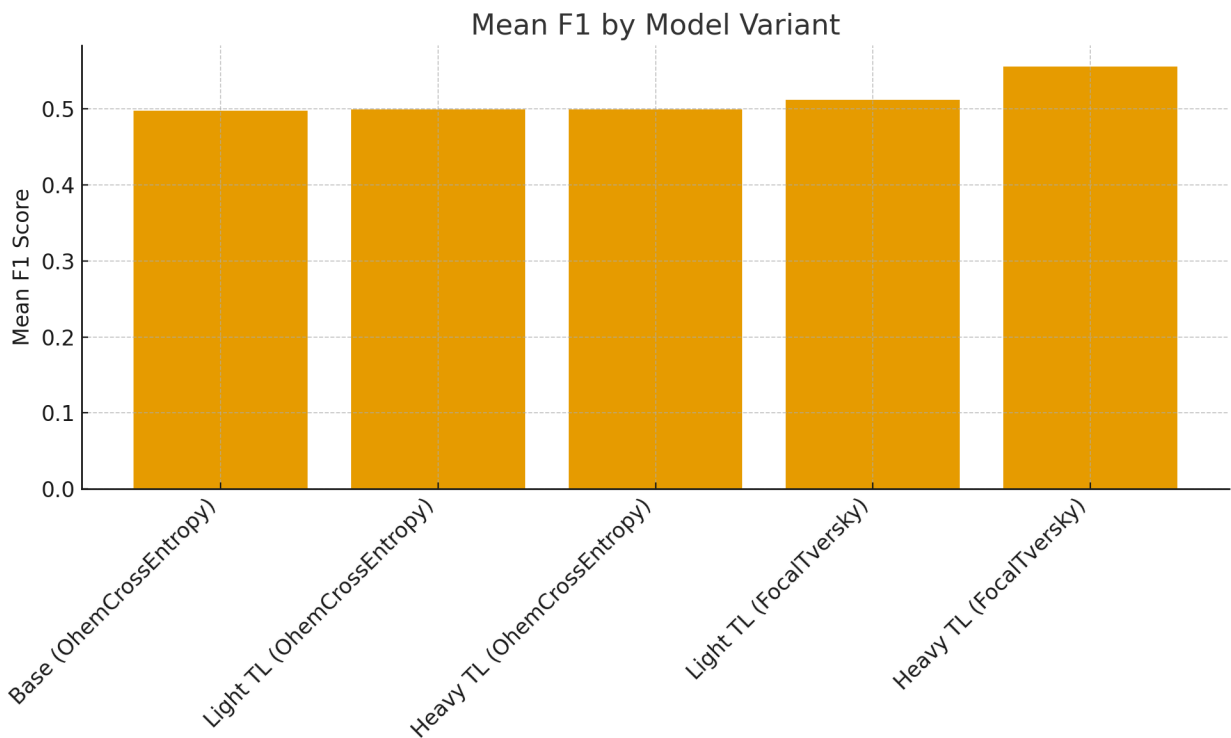
Learning Rate	Iterations	mIoU	IoU_bg	IoU_crack	Precision_bg	Precision_crack	Recall_bg	Recall_crack	F1_bg	F1_crack	mean_Precision	mean_Recall	mean_F1	FPS	Kappa	Dice
0.01	20000	0.5285569	0.996819	0.060295	0.9972731	0.309425183	0.999543	0.06967037	0.998407	0.113733	0.653349154	0.53460679	0.55607	24.53	0.112777	0.55607
0.02	20000	0.5309029	0.99643	0.065375	0.9973174	0.219192254	0.999108	0.08522204	0.998212	0.122728	0.608254838	0.5421651	0.56047	25.08	0.121286	0.56047
0.01	40000	0.5355199	0.996849	0.074191	0.9973214	0.347562255	0.999525	0.08619576	0.998422	0.138134	0.672441819	0.54286021	0.568278	23.8	0.13713	0.568278

Table: FocalTversky base parameters learning rates and iterations comparison

Comparing the results for aggressive vs base parameters for models trained on FocalTversky loss, crack IoU had a moderate but marked decrease compared to the baseline parameters, suggesting that either a less aggressive ratio of α to β or a smaller γ exponent results in improved real world segmentation for SHM.

Learning Rate	Iterations	mIoU	IoU_bg	IoU_crack	Precision_bg	Precision_crack	Recall_bg	Recall_crack	F1_bg	F1_crack	mean_Precision	mean_Recall	mean_F1	FPS	Kappa	Dice
0.01	20000	0.5263168	0.99671	0.055924	0.9972637	0.259780724	0.999443	0.06652428	0.998352	0.105924	0.62852219	0.53298371	0.552138	17.64	0.104855	0.552138
0.02	20000	0.5246412	0.996696	0.052587	0.9972522	0.247423397	0.999441	0.06259937	0.998345	0.099919	0.622337786	0.53102	0.549132	25.25	0.098853	0.549132
0.01	40000	0.535326	0.996708	0.073943	0.9973312	0.296221024	0.999374	0.08970232	0.998352	0.137705	0.646776129	0.54453811	0.568028	23.95	0.136529	0.568028

Table: FocalTversky aggressive parameters learning rates and iterations comparison



Conclusion

Citations

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